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May 13, 2015

Dr. James R. Marsden
Editor-in-Chief
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Dear Dr. Marsden,

Attached please find our paper titled: “Network Decision Support Systems: A Conceptual Model for Decision Support in the Era of Social and Mobile Computing “ (Authors: Bordetsky, A., Dolk, D., and Mullins, S.), which we are submitting for possible consideration in the *Decision Support Systems.*

Please let me know of any questions.

Sincerely,

Alex Bordetsky, PhD
Professor, IS
Fulbright Senior Scholar
Director, Center for Network Innovation and Experimentation

***Biographical Note**

Dr. Alex Bordetsky is tenured Professor of Information Sciences at the Naval Postgraduate School. Professor Bordetsky is Director of the NPS Center for Network Innovation and Experimentation (CENETIX). He is a recipient of prestigious Robert W. Hamming Interdisciplinary Research Award for the pioneering studies of collaborative technologies and adaptive networking and the Fulbright Senior Fellowship Award for the experimental studies of unconventional tactical networking. Dr. Bordetsky is the Principal Investigator for the renowned TNT MIO Military-Academic Experimentation Campaign, which is now in transformation to Maritime-Land CWMD operations study. His research accomplishments are featured in the AFCEA *SIGNAL* Magazine, *Via Sat,* USSOCOM *Tip of the Spear* Journal, *Pentagon Channel, and* Fulbright Newsletter.

Dr. Bordetsky publishes in major IS journals including Information Systems Research, Telecommunication Systems Modeling and Analysis, International Journal of Mobile Wireless Communications, and International Command and Control Research Journal.

Network Decision Support Systems:

A Conceptual Model for Decision Support in the Era of Social and Mobile Computing

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Abstract. We introduce the concept of a mesh network decision support system (NWDSS) emerging from collaborative decision-making environments which originate in ad hoc mobile mesh networking of humans and sensors. The NWDSS is largely an emergent system combining social, technical, and information networks, and as a result, differs substantively from traditional DSS in terms of complexity, decision-making processes, man-machine interactions, and knowledge dynamics. We enumerate a representative list of distinguishing NWDSS properties culled from multiple experiments in maritime interdiction operations conducted by the Naval Postgraduate School. We put these properties in context by presenting a 3-pronged conceptual model consisting of a mobile infrastructure to capture the intricacies of mobile information systems, transactive memory systems for representing the knowledge mesh, or organizational memory, inherent in NWDSS, and emergent decision-making to address the dynamic and agile collaboration processes that arise in volatile, unstructured decision environments. The conceptual model is implemented in principle via network decision support services from which we suggest a variety of research avenues in each of the three areas to develop the NWDSS paradigm more fully. Our contribution is the articulation of a new type of information system that incorporates social media and mobile computing into the landscape of decision support.

Keywords: mobile computing, decision analytics, decision support, emergent systems, collaborative decision making, expert reach back, social networks, transactive memory

1. Introduction

The objective of this position paper is to explore decision support in the context of self-organizing, mobile, socio-technical networks. Building from the Internet era which connected users but with nodes primarily in fixed locations, the conjunction of mobile networks and smart handheld devices have added mobility to nodes enabling the emergence of mobile human sensor nodes, often facilitated via the availability of Apps. The proliferation of social media, both stationary and "on the move" has generated yet another dimension to networked communications, and particularly to decision support in the form of open source decision support environments. Indeed, in many cases, social and mobile computing have created highly dynamic and unstructured decision scenarios, such as the Boston Marathon manhunt, emergency response situations, political upheavals (e.g., Arab Spring uprisings) and various other forms of crowdsourcing, which require new thinking about how to deploy decision support systems (DSS). The increased presence of sensors such as unmanned vehicles in networks adds another degree of complexity requiring the coordination of simultaneous human-human, human-machine and machine-machine communications. Equally demanding high volatility decision-making situations such as emergency response, humanitarian relief, and distributed autonomous unit military operations also challenge traditional DSS thinking while simultaneously presenting opportunities for network-centric approaches. These recent advances in technology have introduced unprecedented degrees of flexibility in Information System (IS) applications while at the same time presenting major challenges to IS design, management and control.

Our goal is to integrate mobile, social networks into the decision support landscape by investigating the confluence of social and technological networks coupled with the need for decision-making "on the fly" in dynamic and volatile environments. We propose a new class of information systems, the Mesh Network Decision Support System (NWDSS), as a means for dealing with the high degree of complexity and uncertainty these environments entail. The motivation to leverage the network metaphor for DSS is the recognition of:

- Emergent vs. top down processes as drivers of complexity
- Critical role of mobile computing and mobile networks in contemporary information and communications technologies (ICT).
- Inclusion of social media into the decision landscape
- Network science as a reference discipline for analyzing both physical and social dimensions
- Holistic vs. individual perspective of decision-making ("the network is the DSS").

We provide a brief history of decision support systems and enumerate the properties of NWDSS. We then propose a conceptual framework for characterizing NWDSS, and finally identify research issues that this model suggests. Our conceptual framework is a tri-partite model of mobile infrastructure to capture the mobile physical network dimension of NWDSS, transactive memory systems as a means of representing the knowledge network, knowledge mesh and associated knowledge flow in decision networks, and emergent decision-making to embrace the social network of agile, unscripted collaboration. The lessons learned during multiple field experiments in Maritime Interdiction Operations (MIO) conducted by the Naval Postgraduate School, help us to illuminate the framework and highlight the distinguishing characteristics of NWDSS. Although our NWDSS concept is motivated primarily by military application studies, we believe the precepts of NWDSS are generalizable to a wide range of applications, and that NWDSS should thus be considered a unique and important new type of information system.

2. DSS Trajectory

During the 40 years since the inception of DSS, the trajectory of DSS evolution may be viewed as encompassing four phases: Individual, Group, Organizational, and Social Network. The earliest instances of DSS were oriented towards supporting individual decision-makers in structured or semi-structured tasks such as financial planning, forecasting, and medical diagnosis that were primarily data-driven, model-driven and knowledge-driven DSS applications geared toward specific decisions and specific decision-makers. Individual DSS (IDSS) continue to be important tools benefiting from advances in data analytics (e.g., data warehouse, OLAP, and data mining), modeling technologies in the form of sophisticated software systems for analytical and computational modeling techniques, knowledge engineering, and visual analytics. Table 1 summarizes in a generalized way the trajectory of each of the major components of IDSS [51] including the network dimension, which was not initially considered an integral part of decision support architecture.

IDSS Component	Trajectory
Data	Relational DBMS→ Data Warehouse → "Big Data"
Models	Structured (optimization) → Process-oriented (ABMS) → OLAP → Predictive Analytics
Knowledge	Expert Systems → Knowledge Engineering → Knowledge Flow → Knowledge Mesh
User Interface	Primitive Graphics → GIS → 3-D (Google Earth™) → Visual Analytics
Network	WAN/LAN → Internet → Mobile Networks

Table 1. Evolution in Components of Individual DSS

Although IDSS are still a recognizable part of the landscape, the focus began to change in the early 1990's with the advent of group decision support systems (GDSS) [42]. In this mode of computer-aided decision-making, multiple stakeholders are enlisted to participate in a synchronous facilitator-driven process using group support software to generate decision alternatives, refine and rank idea these group-derived alternatives, and eventually converge upon a desirable choice. GDSS can be thought of in contemporary terms as an early form of highly structured crowdsourcing, where participants and processes are well-defined a priori. Advantages of this approach include identification of a greater number of alternatives by promoting anonymous contributions during generation, and a greater "buy in" to the eventual decision(s) because of higher group participation. Of more importance to the field of DSS was the realization that there are more and more decisions which involve multiple decision-makers and stakeholders, especially in the team-oriented atmosphere of contemporary management culture, thus GDSS played a key role in expanding the scope of DSS in these directions.

The next logical phase in this progression, organizational DSS (ODSS), resulted from the flattening of organizations made possible by the Internet and the need to operate outside the traditional functional silos both within and across different organizations. This required more collaborative capacity than GDSS were able to provide since the requirements for convening participants in the same physical location rapidly become infeasible in this more general case. As a result, synchronous meetings gave way to asynchronous collaboration as DSS environments shifted from needing to support a group of relatively homogeneous players to a much more diversified heterogeneous set of participants. Emphasis during this phase switched from supporting group meetings to supporting collaboration in all its many guises. In this organizational DSS environment in which collaboration is the most critical requirement, we have multiple decision-makers with multiple (and often incompatible) objectives who are geographically dispersed but who must nevertheless cooperate and coordinate to achieve organizational goals. For the purposes of decision support, this leads to the complex and difficult task of supporting collaborative decision-making in the large, which is a significant increase in scope beyond IDSS and GDSS. One indicator of this difficulty and a sign that collaborative decision-making is a discipline still in its infancy is the lack of a coherent literature on this topic. Also at this ODSS stage, the notion of networks begins to creep into DSS not only from the ubiquity of the Internet and its effects on culture but also from growing awareness of social networks.

The current phase of the DSS trajectory is the class of mesh network DSS NWDSS, which move beyond the narrower perspective of organizations to a broader, more encompassing social dimension. This class of DSS occurs at the intersection of social networking and decision support with the inherent tension between the bottom up nature of social networks and the historically top down approach of DSS. NWDSS transcend conventional organizational structures and environments in the sense that they as often involve decision-making in emergent, ad hoc networked organizations (other terms include "informal networks" or "hastily formed networks") as they do in pre-existing organizational infrastructures. Examples of emergent organizations include the Tweet network which played a significant role in the Arab Spring uprising in Egypt [20], emergency response operations involving multiple heterogeneous agencies, open source projects such as Wikipedia, Linux and Open Science [41], various types of military operations such as search and rescue and maritime interdiction, and of course the multitude of social networking sites which populate the Internet. Tracking the bombers in the Boston Marathon attack is a telling example of the benefits and drawbacks of social media crowdsourcing in an emergency response situation. While the interagency intelligence community was searching relevant data through more or less standard means of decision support using forensic data analysis, they issued in parallel a call to social networking actors, who helped identify the perpetrators and eventually the capture of the second terrorist. However, crowdsourcing in this context also required significant law enforcement resources to sift meaningful content from the flood of messages which bombarded investigators.

Primary technology drivers for the NWDSS version of decision support are the maturation and widespread availability of mobile networks, the advent of smart mobile handheld devices, and the continued refinement and proliferation of sensors as we elaborate in the following section.

3. NWDSS: Next Step in Decision Support

The key differentiator in NWDSS environments is a shift in the unit of analysis from solely decision-makers (whether individual and/or/ group and/or organizational) to networks, both technological and social, which connect the decision-makers. The symbiosis of these two types of networks combined with the increasing interchangeability of human decision-makers and machines and the need for collaboration among highly heterogeneous agents leads to extreme, complex and volatile decision-making environments which are situated much further towards the unstructured end of Simon's decision spectrum [49] than has historically been the case in the earlier phases of DSS. Below we enumerate salient properties of NWDSS which distinguish it from the more traditional forms of DSS.

- *Fluid, multimodal networks.* This is perhaps the signature aspect of NWDSS, namely in the most general case, a mobile network with the objective of decision-making and consisting of highly heterogeneous nodes entering and leaving the network in unanticipated fashion. Nodes in an NWDSS may be humans, sensors, and/or software agents. Human nodes may represent individuals as well as different organizations such as military units, law enforcement agencies, firefighters, first responders, non-governmental organizations (NGOs), and various local, state and federal governmental agencies. The human nodes typically include, but are not confined to, the decision-makers in the network although intelligence in the network in the form of software agents may subsume lower level operations-oriented decisions such as network monitoring and management.

  Networks and sub-networks may emerge or dissipate in ad hoc fashion as various nodes join or leave the community so that the structures of the technological, social and information networks are dynamic and unpredictable in the most general case. Instances of a node leaving the network may be a person who turns off his cell phone, a sensor which has become disabled or gone into silent mode, or a node which falls outside the range of the mobile network either because the node or the network moved. The associated semantics of missing nodes and associated node continuity and network stability is a challenging research objective of NWDSS.
- *Multidimensional technical, social, and information networks.* NWDSS environments consist of technical networks (e.g., Internet and mobile networks), social networks of connected human players and machines (see below), and information networks governing information and knowledge flow, all operating in parallel and in the service of decision-making. NWDSS thus transcend the simple one-dimensional physical model of networks to incorporate the human dimension in the context of decision support. This offers a robust situation for the synthesis and synergy of the corresponding modes of network analysis. For example one might think of decision-making roles as corollaries of physical network nodes such as routers and gateways, examine physical networks through the lens of social network concepts such as degrees of separation and small worlds, and consider error propagation in social as well as physical networks.

- *Sensor-rich.* NWDSS in the most general context may contain non-human and human agents acting as sensors. The proliferation and miniaturization of non-human sensors is a major driver in the emergence of NWDSS, particularly in the arena of emergency response, homeland defense and military tactical operations. For example, unmanned vehicles (air, ground and sea) play a pivotal role in search and rescue, border patrol, and threat detection operations. As sensors become smaller and smaller, their scale covers several orders of magnitude from satellites to the level of nano-sensors embedded in human agents. The rapid evolution of smart mobile technology accelerates the sensor-based nature of NWDSS since mobile users themselves can act as sensors with the means to transmit and receive rich information in many different modes in very near real time. Coordinating a portfolio of sensors with a potential of high scalability renders data fusion a major issue for any meaningful decision analytics.
- *Multi-modal communications: Simultaneous human-machine, machine-machine and human-human interactions.* The presence of unmanned system-based sensors in the network extends significantly the requirements of collaborative activity beyond just human-to-human interaction. Traditional DSS focused primarily upon human-computer interactions whereas NWDSS environments may include an array of sensors, some driven by the unmanned aerial, surface, or ground systems, some being geographically distributed as unattended sensors, having human experts operating the mobile sensor infrastructure remotely. Man-to-machine communication is critical, for example, when there are unmanned vehicles which must be guided to useful locations. Machine-to-man communication is necessary when an autonomous sensor detects an event of interest. Machine-to-machine communication may be required to coordinate an unmanned aerial vehicle, for example, with a ground based camera as in border patrol applications. Human-to-human interaction can occur at many levels, for example two individuals communicating by mobile phones, teams negotiating or sharing knowledge across the network, or military operators in the field relaying information back to their headquarters. How these new simultaneous *human-machine, machine-machine and human-human interactions* differ from the classic DSS environment is in the high degree of *autonomy* in

communicating sensing, planning, and execution of the decisions made. For example, remotely located decision-makers operate unmanned autonomous systems by submitting interim navigation points to task the autonomous system through highly discrete moments of time [11]. Correspondingly, understanding and implementing different levels of autonomy becomes an integral part of NWDSS processes. All interactions are critical in generating and sustaining *situational awareness* of the environment in which the NWDSS is operating.

- *Fluid decision-making roles.* Not only are the players in the network fluid, but the roles they perform in the decision-making process may be as well. Depending upon the context, an individual may concurrently be responsible for simply passing information along the network (repeater), deciding who should receive what information (router), or "translating" expert knowledge into actionable knowledge for other participants (gateway). Sensors such as unmanned vehicles, for example, may operate as information transmitters in one mode, and autonomous decision-makers in another.
- *Collective intelligence: "knowledge mesh" and emergent knowledge processes.* Mesh network decision support environments involve many different players and agents who form a virtual team for the purpose of achieving a set of objectives (e.g., solving a mathematical theorem, building an operating system, providing effective humanitarian relief, containing a fire, monitoring and interdicting narcotics traffic, search and rescue). The knowledge contained within this cross-organizational network is a critical parameter in the efficacy and effectiveness of the decision-making required to achieve the group objectives. In knowledge management terms, each network can be thought of as a *knowledge mesh,* embodying in principle the collective knowledge base of each node, coupled with *emergent knowledge processes (EKP)* which continually act upon the knowledge mesh by enhancing or otherwise modifying knowledge flow between the knowledge bases.
  EKP systems are characterized by a changeable cast of users whose work profiles cannot be identified a priori, emergent work processes that result in fuzzy initial requirements that only become crisp through a succession of functional prototypes and a strong emphasis on knowledge flow across a network [36]. One of the key knowledge flow processes in NWDSS is that of *expert reach back,*

wherein decision-makers require access to a specific knowledge base (whether human or artificial) as part of their decision-making process. For example, in a maritime interdiction operation, a ship boarding team may require real time confirmation from a nuclear expert (human node in the network) about whether a discovered device onboard (perhaps a centrifuge) is intended for use in a weapon. In a similar vein, biometric data gathered in the same operation may be matched against existing databases (artificial agents in the network) to identify individuals who are potential threats. We suggest that this aspect of expert reach back argues for a transactive memory system approach to organizational knowledge which we discuss in more detail as part of our conceptual model.

- *Agile, collaborative decision-making "at the edge".* Mesh network decision support environments such as emergency response and tactical battlefield operations exhibit a pronounced shift toward unstructured decision-making situations. These tend to be chaotic, unscripted, and fraught with high risk and severe time pressures in which decision-making must frequently be done with highly imperfect information and without a clear delineation of the available alternatives, the possible outcomes, or the probabilities attached to each. Furthermore, the norm typically involves many decision-makers from disparate organizational agencies who must collaborate and coordinate on series of interrelated decisions.

We claim the properties enumerated above and summarized in Table 2 comprise a representative, if not exhaustive, depiction of NWDSS environments. In the next section we develop a conceptual model to accommodate this new form of DSS, and to establish a research agenda for the many complexities that these environments entail.

NWDSS Property	Description
Fluid, heterogeneous nodes	A NWDSS is a mobile network with the objective of decision-making and consisting of highly heterogeneous nodes entering and leaving the network in unanticipated fashion.
Multi-dimensional: Complementary technical,	NWDSS environments consist of *technical networks* (e.g., Internet and mobile networks), *social networks* of connected human players and

social, and information networks	machines (see below), and *information networks* governing information and knowledge flow, all operating in parallel and in the service of decision-making.
Sensor-rich	NWDSS in the most general context may contain non-human and human agents acting as sensors ranging from nano to macro in scale.
Multi-modal communications	The presence of unmanned system-based sensors in the network extends collaborative activity requirements beyond just human-to-human interaction to include human-to-machine and machine-to-machine communications.
Fluid decision-making roles	A specific node may concurrently be responsible for simply passing information along the network (repeater), deciding who should receive what information (router), or "translating" expert knowledge into actionable knowledge for other participants (gateway).
Collective intelligence: "knowledge mesh" and emergent knowledge processes	Organizational, or collective intelligence, is a *knowledge mesh*, embodying in principle the collective knowledge base of each node, coupled with *emergent knowledge processes (EKP)* which continually act upon the knowledge mesh by enhancing or otherwise modifying knowledge flow between the knowledge bases.
Agile collaborative decision-making "at the edge"	Mesh network decision support environments such as emergency response and tactical battlefield operations exhibit a pronounced shift toward unstructured decision-making situations and push decision-making to the "edge" of organizations, i.e. to agents closest to the actions.

Table 2. Properties of NWDSS

4. Conceptual Model for NWDSS

Decision support involves the intersection of organizations and information technology for purposes of decision-making. From the network-centric view, both the organizational and technological dimensions rely upon networks: primarily social and knowledge networks in the former, and mobile networks in the latter. We expand upon this simple model to propose a conceptual framework for NWDSS as shown in Figure 4.

The broad overarching context for our model is *digital infrastructure* as characterized in [14, 52, 56]:

> "*Digital infrastructures* can be defined as shared, unbounded, heterogeneous, open, and evolving sociotechnical systems comprising an installed base of diverse information technology capabilities and their user, operations, and design communities." [14].

"… digital infrastructures herald a new stage in the evolution of IT, reflecting the fact that IT has become deeply socially embedded, (and) is coordinated through diverse sociotechnical worlds and numerous standards ..." [52].

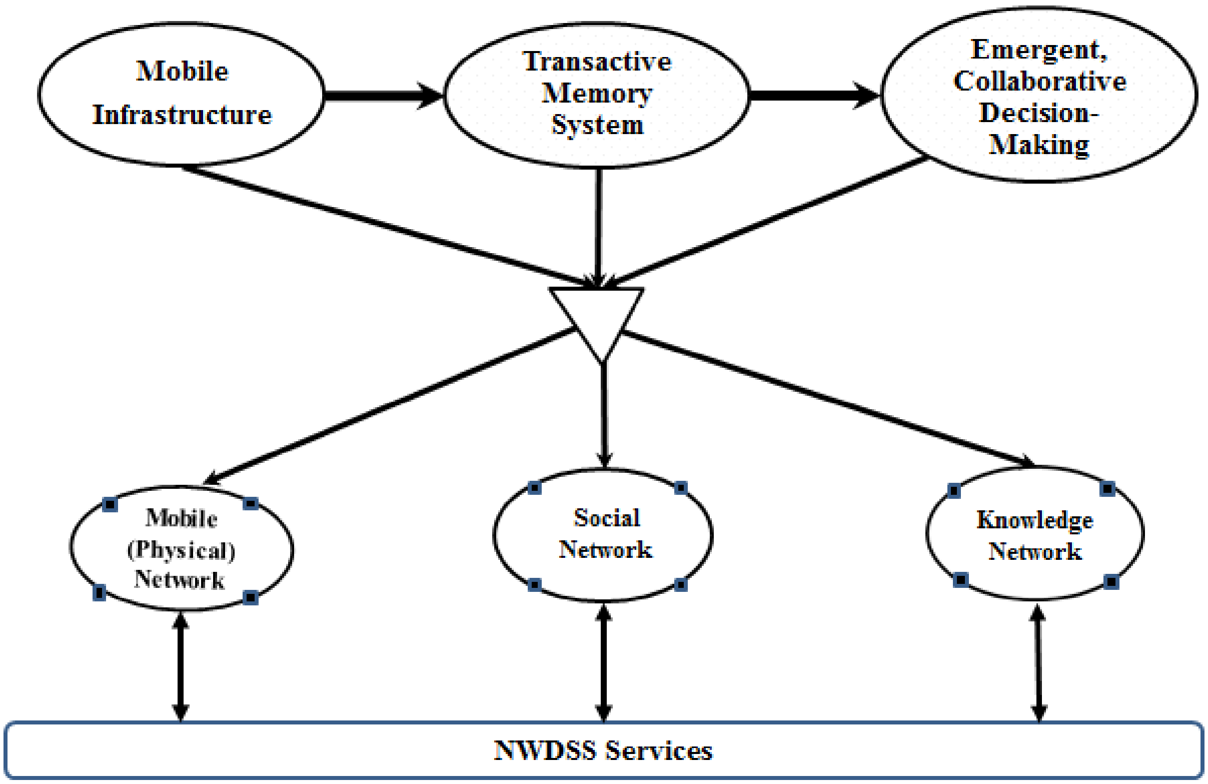

Figure 2. Conceptual Framework for Network DSS

The above definitions and characterizations of digital infrastructure align very well with the elements of NWDSS we enumerated in Section 3. NWDSS are highly heterogeneous shared environments that combine social and technological networks as we have described. The emergent knowledge processes which characterize NWDSS prevent the a priori specification of requirements, functions, and applications satisfying those requirements. Rather, these evolve as the system emerges. NWDSS are open and unbounded in this sense as well, which we see in the use of wireless mobile networks and smart phones in social media settings.

Although *digital infrastructure* is a term which has been in use for a couple of decades, [56] argues that it constitutes a fruitful area of research which has largely been ignored by the IS research community. We contend that NWDSS constitutes a new form of DSS which is an exemplar of digital infrastructure. With this in mind, we adopt digital infrastructure as a conceptual superset which parallels this evolution, but which needs to be refined to accommodate the mobile network-centric focus of NWDSS.

We employ three major conceptual components in our mobile-oriented enhancement of digital infrastructure to capture the complexity and dynamics of mesh network decision support environments: *mobile infrastructure* to address the physical mobile network, *transactive memory* to represent the knowledge network (mesh), and *emergent collaborative decision-making* to capture the social network aspects of decision support. The architectural dimension underlying these three components is a portfolio of NWDSS services. We discuss each of the pillars of our conceptual model in turn.

4.1 Mobile Infrastructure

We use mobile infrastructure in the context of network infrastructure which is, in turn, a subset of overall enterprise IT infrastructure. Our treatment of mobile infrastructure focuses upon the physical dimension of mobile networks, namely the hardware and software resources that enable mobile network connectivity, communication, operations and management. These issues are complex in the NWDSS world because not only are the network nodes mobile, but networks themselves are not necessarily fixed but may very well be "on the move" as is the case with emergency response, humanitarian relief, and many types of military operations. The major challenge in mobile networks is maintaining connectivity which, in turn, is necessary, although not sufficient, to support situational awareness. We identify three major research and management challenges with respect to mobile infrastructure and NWDSS: adaptive networks, embedding DSS functionality in network nodes, and the management of control and change within the networks.

Adaptive network operations. Ad hoc mobile networks are essential components of emerging infrastructure environments, however as we have experienced in the MIO experiments, they have proven to be fragile, requiring constant monitoring and control to deliver acceptable levels of performance. From a management point of view, incorporating adaptive behavior at the network operations level is not only desirable, but necessary, to address network management issues at the socio- and technological levels [9]. Some of the problems facing “on the move” mobile networks include maintaining situational awareness and network connectivity in real or near-real time when the boundaries of network coverage may be fuzzy, when nodes themselves are moving and may go out of range (or disappear for other reasons, e.g. faulty communications equipment), when secure communications are compromised, and a host of other possible disruptive events.

Incorporating DSS functionality in the physical network. One approach to implementing adaptive network management is the ability to embed some level of decision support in the physical network itself. For example we are currently experimenting in MIO with embedding network monitoring DSS functionality into the network via the introduction of *self-aware hyper-nodes*, These higher level nodes are based on 8th layer enabled hyper-nodes [7], which contain minimal elements of network operation center functionality and are capable of negotiating video, text, and sensor data services with their neighbors. This self-organizing, disruption-tolerant network operation architecture has the potential to maintain application service continuity across the nodes that generate and receive data over disjoint moments of time in different locations that are not initially connected.

Hyper-nodes are but one mechanism for promoting adaptive behavior but a more comprehensive model is needed. A major research challenge of NWDSS in the mobile infrastructure sphere is thus developing a theoretical model of network adaptation which can be tested experimentally, and which integrates critical network systems phenomena such as feedback, open (generative) systems, as well as the traditional closed hierarchical systems familiar to IS research.

Network control vs. flexibility. Closely related to network adaptation, at both the physical and social levels, is the tradeoff that must be considered between network control and flexibility [56]. Truly ad hoc networks often have few, if any, control mechanisms initially but may evolve them as needs dictate (the Linux open source programming community is a good example). We characterize this in terms of situational vs. context awareness wherein we mean *context awareness* to signify a particular node’s perception of its immediate environment, and *situational awareness* to signify a more integrated command and control perspective of the overall environment the network is sensing. (The distinction from a decision-making perspective may alternatively be thought of in terms of individual vs. organizational DSS.) The integration of context and situational awareness also raises many interesting research issues around the various kinds of interaction dynamics (expert ←→ nonexpert; expert ←→ sensor; nonexpert ←→ sensor; sensor ←→ sensor) which exist in sensor-based mobile networks. The interplay between these two modes will eventually determine the level of control the network exhibits and is strongly related to the existence and flow of knowledge within the network (what we term the *knowledge mesh*).

4.2 Organizational memory and collective intelligence via transactive memory systems (TMS)

Whereas we use mobile infrastructure as the medium to capture the technological aspects of NWDSS, we now turn our attention to the social network segment which we envision as being effectively a knowledge infrastructure within the mobile mesh infrastructure. Mesh network decision support environments involve many different players and agents who form a virtual team for the purpose of achieving a set of objectives (e.g., containing a fire, monitoring and interdicting narcotics traffic, search and rescue). The knowledge contained is a critical parameter in the efficacy and effectiveness of the decision-making required to achieve desired objectives. As described in Section 3, we characterize structural and process components of this infrastructure as *knowledge mesh* and *emerging knowledge processes* respectively. There are several dimensions of knowledge mesh in NWDSS: the social or organizational memory resident in the network's human-to-human communications, human-to-machine interactions, and machine-to-machine protocols.

Knowledge processes acting upon the NWDSS knowledge mesh have as their overarching functional objective the creation and maintenance of persistent *situational awareness* to support decision-making. Specific processes tend to emerge in volatile environments but we can nevertheless identify generally useful functions, for example, information gathering and fusion, sensemaking and "mental model" creation, task identification, and task execution. Functionality depends in part upon the decision-making models employed, a topic we address in detail in the next section, but the key requirements from a knowledge standpoint as shown in Figure 3 are awareness of:

- what knowledge exists where in the network (meta-knowledge),
- who needs what knowledge and when (actionable knowledge requirement), and
- how to provide (i.e., access, translate, and relay) that necessary knowledge to those who need it (knowledge flow).

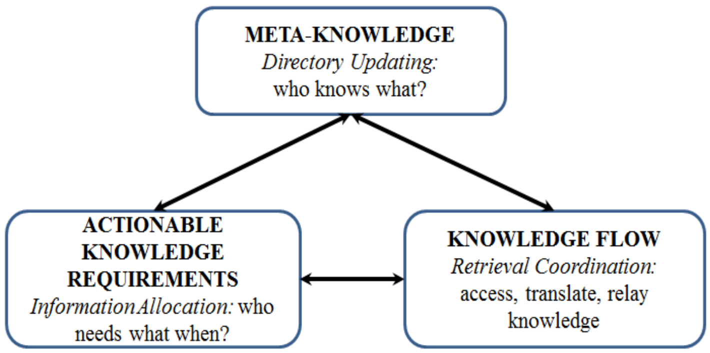

Figure 3. Elemental Components of Transactive Memory Systems

The knowledge requirements above align very well with the characteristics of a *transactive memory system (TMS).* The concept of TMS was first introduced by [59] and defined as “a system through which groups collectively encode, store, and retrieve knowledge”. Early TMS research initially focused primarily on individual and dyadic personal relationships. [60] reframed TMS in the context of networks identifying three associated elemental processes: directory updating, information allocation, and retrieval coordination, which correspond informally to knowing who knows what in a group, assigning memory items to specific group members and accessing, or retrieving, knowledge by leveraging who knows what. The applicability of TMS has gradually expanded since then to encompass groups, larger collectives, organizations, and recently social networks, with interest in TMS research growing accordingly.

Two recent surveys note that the conceptualization and objectives of TMS have become more diffuse during its evolution from individuals to collectives, and argue for a reassessment and recalibration to reflect the "scope creep" of TMS research (Table 3). These enhancements help bring TMS and NWDSS closer together but current TMS research still envisions a much more stable organizational milieu than we typically expect to encounter in NWDSS applications. [35] enumerates in detail several areas where current assumptions made by TMS theory must be relaxed and/or altered to address the case of emergent response groups.

Source: Ren and Argote [46]	*Source: Lewis and Herndon [32]*	*Source: Majchrzak, Jarvenpaa, Hollingshead [35]*
TMS as a multidimensional construct	Definition of TMS in terms of • differentiated as well as shared knowledge, • process as well as structure, • dynamic processes involved in creating new collective knowledge.	Coordinate knowledge processes in the absence of a shared meta-structure. There likely will be few, if any a priori knowledge meta-structures, so mechanisms for coordinating emergent knowledge processes must evolve "on the move".
Consistency in measuring TMS	Reinterpret relationship between TMS concepts and TMS measurements	Replace credibility through expertise with trust in action. Trust and credibility are not deep structures in ERGs, rather they are more likely to emerge through useful task-based action as much as through knowledge brought to bear.
Organizational level TMS; Building strong TMS at group level; Linking groups via small world networks;	Integrate TMS and Task Type via Task Processes, Structural Qualities, Relevance (TMS and group performance); and Development; Incorporate social media into the TMS landscape	Tailor the role in expertise. Expertise may need to be de-emphasized and complemented by a group's capability and motivation.
Integration of theory-driven and practice-driven TMS; Full-cycle research between field observations to formulate hypotheses / models and experiments to test same (see Figure 2 in Section 3)	Computational modeling to generate hypotheses that can be tested in laboratory and field settings	Extending TMS theory to emergent response groups where related knowledge processes and structures are dynamic, fuzzy, and volatile.

Table 3. Extensions to TMS

The correlation between a DSS knowledge mesh and TMS is apparent and underlies our motivation for including TMS as the second leg in our conceptual model. However, we are particularly interested in the intersection of transactive memory and social network environments, and this is an area which requires significant extensions to current TMS theory and application. We believe even further enhancements of TMS are necessary that explicitly incorporate concepts from both technological and social network concepts. For example, we have found from the MIO experimentation [9] that different nodes may play different roles in the decision-making process that mimic the functions of nodes in a physical network. An agent may serve as the equivalent of a router broadcasting information at one point in time and later as a Gateway, translating knowledge from experts to forward operators. Sensors (machines) may simply relay data to the operations center in one mode but coordinate other sensors autonomously in a proactive mode. This leads us to extend Wegner's computer-based model by considering the roles of various agents (including decision-makers) as counterparts of the physical components of a network (Table 4). We note that roles may very well emerge during the course of events, especially in ERG conditions, so that one node may end up serving several functions not all of which may have been anticipated, and further, as nodes leave the network either voluntarily or otherwise, roles may temporarily disappear. For example, if an expert collaborator node suddenly falls outside the range of the mobile network, that expertise will be unavailable to the knowledge mesh until connection is reestablished.

Component	Physical Network Function	Knowledge Flow Function	MIO Example	TMS Function
NWDSS Repeater	Relays messages from one node to another possibly at higher power to extend range of network but without modifying content	Relays information from one player to another without modification	Small unit human relay	Information allocation
NWDSS Bridge	Connects multiple network segments	Expert-to-Expert Collaboration	Joint spectra analysis & emerging detection services	Access and retrieval
NWDSS Router	Traffic director sending packets to next node in network	Decides who should receive what information	Experts exchange messaging & file uploads with Combat Swimmers Unit Leader who routes message to one of the swimmers for subsequent action.	Information allocation; Directory updating

NWDSS Gateway	Network node equipped for interfacing with another network that uses different protocols.	Translates info / knowledge from specific player(s) to other player(s); Cross-functional roles translator	Opns center manages knowledge flows between swimmers and experts on adjudication of target vessel	Access and retrieval

Table 4. Network Role Model Counterparts for NWDSS Knowledge Flow

It would also seem apparent that the underlying network structure(s) of a TMS would be a significant determinant in the effectiveness and performance of a group. Thus, knowing whether a network was random or small world or scale-free would be useful in determining even *post facto* whether certain structures were more or less effective in generating robust TMS. Other social network characteristics such as degrees of separation, connectedness, closeness, and 'betweenness' of nodes are also relevant here. Closeness, for example, which is a measure of the overall degree of separation of a node from all other nodes, may be an indicator of a node with better situational awareness when its value is low. Being able to identify such nodes and ascertain whether their situational awareness was indeed superior would be a valuable outcome for TMS research. A node with high 'betweenness', i.e. a node which connects different communities, has potentially great influence over what flows in and out of the network and may very well serve concurrently all four roles shown in Table 4. With respect to network reliability, knowing which nodes are highly connected (hubs) and which are key 'between' nodes is critical in determining potential single points of failure should those nodes become inoperative during the course of operations. Finally, being able to view social networks through different lenses such as knowledge flow or trust or power can also enlighten TMS research. [26] present a case study which shows how charting an advice social network in a specific organization reveals clearly where expertise resides therein. The graph of the corresponding trust network, however, reveals an entirely different network structure.

Our impression from surveying the literature on organizational memory and TMS in particular, is that there is sparse overlap with network theory per se. Once we adopt the view of organizational memory as a network, not only can we assimilate the basic structural concepts described above, but we can begin to think of organizational memory in a more holistic manner. Thus, perhaps it makes sense in some contexts to view organizational memory as a neural network with excitatory and inhibitory nodes and processes such as back and forward propagation in effect. Such an approach brings into play a vast body of neural net research with accompanying powerful analytical capabilities. In a similar vein stretching the imagination even further, we can also begin to think of organizational memory as a single coordinated consciousness or intelligence, adopting metaphors from neuroscience and brain science in the process. Adopting this analogy, [8] explicitly consider a TMS as corollary to the human cortex. As neuroscience continues to make huge strides in our understanding of the brain's processes, for example how long term memory is "stored" in the brain [23], it is not unreasonable to expect that such research will also eventually enlighten us about how certain aspects of decision-making take place at these micro biological levels. Although we must be very cautious in applying low level biological and physical phenomena directly to macroscopic human behavior, research on organizational memory and TMS could someday help facilitate a convergence of sorts between the biology of decision support and the largely artifact-driven discipline it is today. We believe that NWDSS research can play an integral part in this rapprochement.

TMS in concert with mobile infrastructure constitutes a conceptual framework which integrates the social and technological dimensions of dynamic organizations which characterize NWDSS. Our network-centric extensions to TMS above are primarily structural in nature and do not significantly address the process dimension of TMS. What remains to be discussed to address this are the decision-making processes which must be supported by this confluence of social and technical domains.

4.3 Emergent, collaborative decision-making

4.3.1 Traditional decision-making models in DSS

The decision-making models typically referenced in the DSS literature (Table 5) are too limited to apply in full to NWDSS environments. Many of the earlier models such as Simon's Intelligence / Design / Choice [49] and Boyd's OODA (Observe / Orient / Decide / Act) loop [43] are focused upon individual decision-making whereas contemporary networked environments are much more likely to be group- or team-centric where there are typically multiple decision-makers requiring significant collaboration, often in near real time circumstances.

Decision-Making Model	**Description**
Static Models	
Neo-classical economics [48]	Perfect information of alternatives, outcomes and utility function resulting in optimization of expected utility.
Bounded rationality [50]	Satisficing through heuristic search rather than optimization
Simon's problem-solving [49]	4 phases: 1. Intelligence (problem identification); 2. Design (model-building); 3. Choice; 4. Implementation
Bayesian [44]	Identification of all alternatives and outcomes with a priori and a posteriori probabilities assigned
Decision analysis: multi-criteria decision models [e.g., 47]	Identification of all decision criteria, associated yield functions and weights yielding ranked list of decision alternatives.
Dynamic Models	
OODA [43]	4 phases with rapid feedback loops: 1. Observe (situational awareness); 2. Orient (identify courses of action-COA); 3. Decide (select COA); 4. Act
Naturalistic or Recognition Pattern-Directed [25]	Decision-makers rely upon case-based templates derived from previous experience to identify situations and make decisions accordingly
Group Support Systems (GSS) [42]	Synchronous decision room with a facilitator, using software designed for brainstorming, ranking and voting to evaluate group-generated alternatives for a single decision.
Robust decision-making [57]	Non-probabilistic approach to decision-making in situations of high uncertainty; *robust* means best decision reached under given (fuzzy) conditions regardless of outcomes.

Table 5. Decision-making Models in DSS

We argue that such models must be based upon a paradigm of emergent decision-making in keeping with the emergent nature of network-based complexity. Collaboration in the context of emergent decision-making is qualitatively different, however, from that which is typically discussed in the DSS literature. DSS-based collaboration research emanated initially from work in group support systems (GSS) which relied strongly upon skilled facilitators to attain effective decision-making outcomes [42]. As thinking evolved beyond the GSS paradigm to more asynchronous modes of collaboration, research in collaborative environments nevertheless adhered to traditional IS design principles with emphases on identifying *a priori* objects and processes (e.g., goals, tasks, roles, products, activities, procedures, etc.) as overarching requirements for collaborative design science.

A priori does not work, however, in the emergent climate of NWDSS. Collaboration in firefighting or humanitarian relief assistance is inherently different than collaboration on a software project, for example. In the latter, there is ample time for planning and identifying tasks, roles, activities, deadlines and the like which can be configured into a structured collaboration environment, whereas in emergency response situations, there may be little or no time for pre-planning and the deliberate configuration of resources. NWDSS do not always have a centralized node, or operations center, particularly when emergent networks are involved. Decision-making in NWDSS is therefore pushed out to the "edge" of an organization, i.e. away from centralized command and control and towards the individuals "in the field". This places a much heavier reliance upon *agile* collaborative decision-making which can be thought of as collaborative, or emergent, decision-making under tight time constraints and imperfect information. This advanced level of collaboration is seen as the highest level of a command and control (C2) maturity model (C2MM) used in the software industry and now used widely for assessing process maturity. We note that while C2 environments are of particular interest in our case study, they comprise only a subset of all NWDSS applications; we find the C2MM interesting because it explicitly addresses, although in very general form, the collaboration requirements of what we call emergent decision-making.

In summary, collaboration in NWDSS environments is much more likely to be ad hoc than scripted, bottom up rather than top down, and "plan resistant" rather than "plan friendly". A further complicating factor in NWDSS involving sensors and unmanned vehicles is that collaboration must now consider man-machine as well as man-man interactions as part of the overall decision-making process. How then does one address decision-making in such a turbulent networked environment with multiple players, multiple decisions, and multiple decision-makers?

4.3.2 Sense-Analyze-Adapt-Memory (SAAM) decision-making process

We begin by adopting the Cynefin framework (Table 6) which prescribes different sequences of actions contingent upon the type of decision environment encountered [30]. This framework constitutes a more meaningful refinement of Simon's conventional characterization of decision support environments as structured (Simple), semi-structured (Complicated) or unstructured (Complex; Chaotic) [49] and encompasses the kinds of environments in which NWDSS are likely to exist.

Decision-Making Environment	Associated Activities	Description
Simple	Sense-Categorize-Respond	Cause and effect obvious; Apply *best practices* to *Sense – Categorize – Respond*
Complicated	Sense-Analyze-Respond	Cause and effect requires analysis and/or expert knowledge; Apply *good practice* to *Sense – Analyze – Respond*.
Complex	Probe-Sense-Respond	Cause and effect unpredictable and can only be discerned post facto; Apply *emergent practice* to *Probe – Sense – Respond*
Chaotic	Act-Sense-Respond	Cannot ascertain cause and effect; Generate *novel practice* by process of *Act – Sense – Respond*

Table 6. Cynefin Framework of Decision Environments and Associated Actions [28]

To address decision-making in such a turbulent networked environment, we suggest a basic decision feedback loop model of Sense-Analyze-Adapt-Memory (SAAM) (Figure 7) which includes the transactive memory dimension and can be adapted to any of the Cynefin scenarios. The SAAM model is particularly relevant for tactical networking settings such as the MIO but we believe it is sufficiently generalizable to apply to other networked infrastructures. In this model, decision-makers and stakeholders in the Sense phase continuously receive data and information from sensors and humans in the network which is "stored" in Network Memory as part of the Transactive Memory infrastructure and processing. This information is analyzed and integrated during the Analyze phase as part of the sensemaking process into a persistent situational awareness scenario. Sensemaking in this concept transcends Weick's concept [61] in that it must deal more with ad hoc organizations overlaid upon existing organizations, coalitions, and non-aligned individuals. The Adapt phase is the decision-making part of the process wherein decision-makers decide how to react (if at all) to the situational awareness profile by devising one or more courses of action to be pursued. The timing of this decision loop may be seconds or minutes as in a combat engagement, or days, weeks, or months in a less time-sensitive environment such as a software design project.

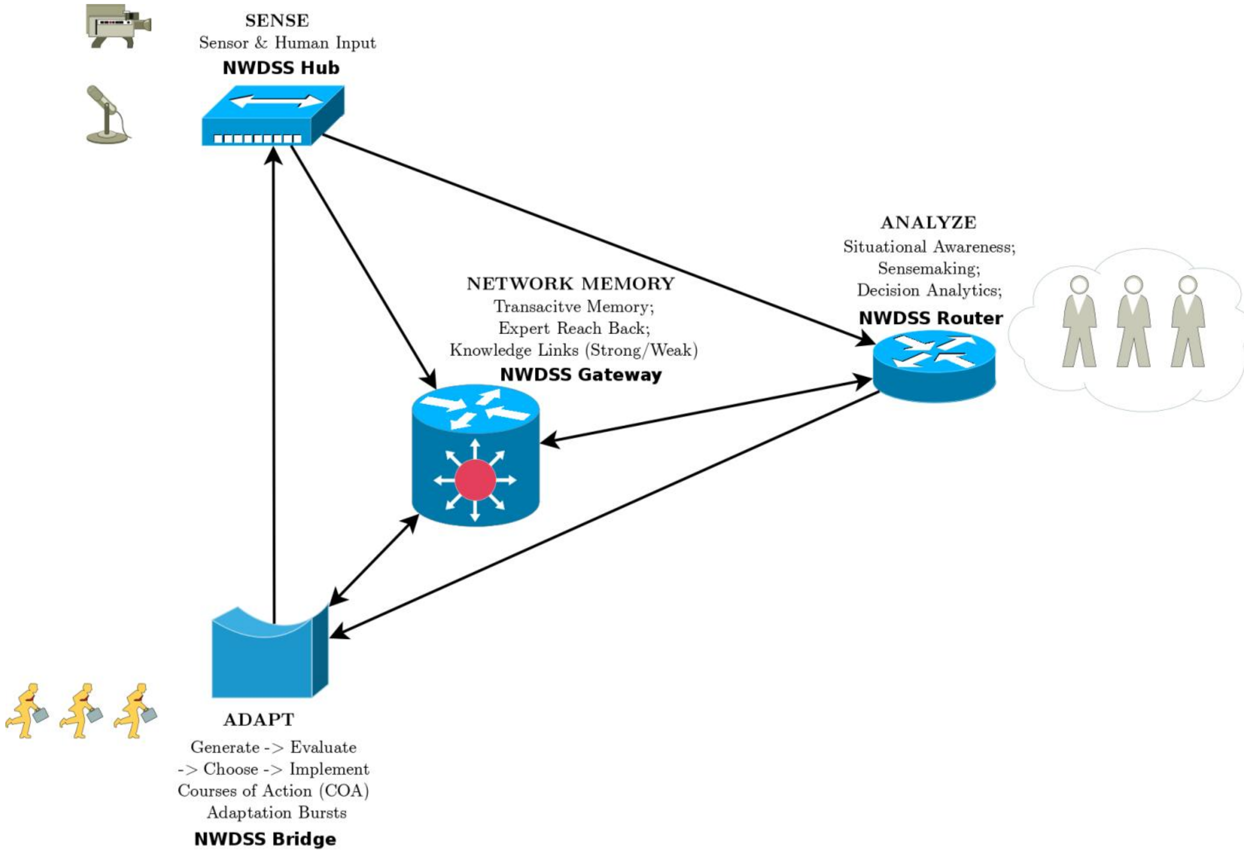

Figure 4. Emergent NWDSS Roles in Execution of Sense-Analyze-Adapt-Memory-Based Decision Cycle Model

Although Figure 4 would seem to correspond most closely to Cynefin's "Complicated" scenario, our notion of "Sense" subsumes "Probe" since sensors such as unmanned vehicles, for example, do exactly that, and our "Adapt" stage is a counterpart of the Cynefin "Respond" activity. Given this interpretation, the SAAM model actually can be adapted to all the various Cynefin environments with minor modifications based upon the starting point of each. The key difference in our framework, however, is the Memory component which uses transactive memory structures and processes to address the knowledge and social network aspects of the decision-making environment

The SAAM decision-making process is likely not only taking place at many different locations and levels within the network, but may also be incorporating other models in the process. Using the battlefield example, soldiers on the ground searching for improvised explosive devices (IEDs) operate in a very time-constrained version of the loop wherein experienced soldiers may rely upon something similar to Klein's case-based decision model [25] as their version of the Adapt process. Decisions at the command and control node(s) may be different, however. Time constraints may not be quite as pressing as on the ground, but the situational awareness will likely be broader and more complex, perhaps requiring coordination of several simultaneous operations. Thus, it would appear that hybrid models of individual and group processes are likely to be present in agile, collaborative decision-making depending upon the location of the decision-maker in the network (edge or center) and the roles they play in the decision process.

The SAAM loop suggests to us that decision-making in NWDSS may be a continuous spiral of emergence and convergence where the Sense-Memory stages represent the emergence of data, information, knowledge and ideas, and Analyze-Adapt represent the convergence of this knowledge into plans of action. This is reminiscent of the Explicit-Implicit Interaction theory [17] which characterizes creative problem solving in complex and ambiguous situations as an alternating current of implicit and explicit processing that ultimately converges to a "solution" or decision. Consider as an example the GSS model of brainstorming, ranking, and voting where brainstorming is the emergent process of identifying ideas and options, and ranking the options is the convergent process resulting in an eventual decision. Emergent processes in networks result in the self-organizing, convergent creation of hubs whether in identifying leaders as knowledge organizers and experts in open source and open science projects or recognizing control nodes where enough knowledge has coalesced to accelerate decision-making (e.g., expert reach back).

4.3.3 Decision Analytics in Near-Real Time

Decision analytics is a critical part for understanding NWDSS processes and functions such as monitoring physical network operations, tracking social network structures, identifying communications patterns, and predicting user behaviors and future events. What distinguishes NWDSS decision analytics from conventional model management is the need for real or near-real time predictive capabilities. For example, adaptive network management in the digital infrastructure requires dynamic operations research models and simulations that can respond quickly to changing situations. This changes the traditional landscape of model management drastically. We cannot afford the overhead of developing and testing traditional static models over long time horizons. Rather models must now be viewed in the context of tightly time-constrained feedback loops requiring them to adapt rapidly to the substantial real-time data streaming that network environments generate [28]. This *model streaming*, or *adaptive modeling*, aligns NWDSS to leverage and contribute to advances in Big Data and data analytics.

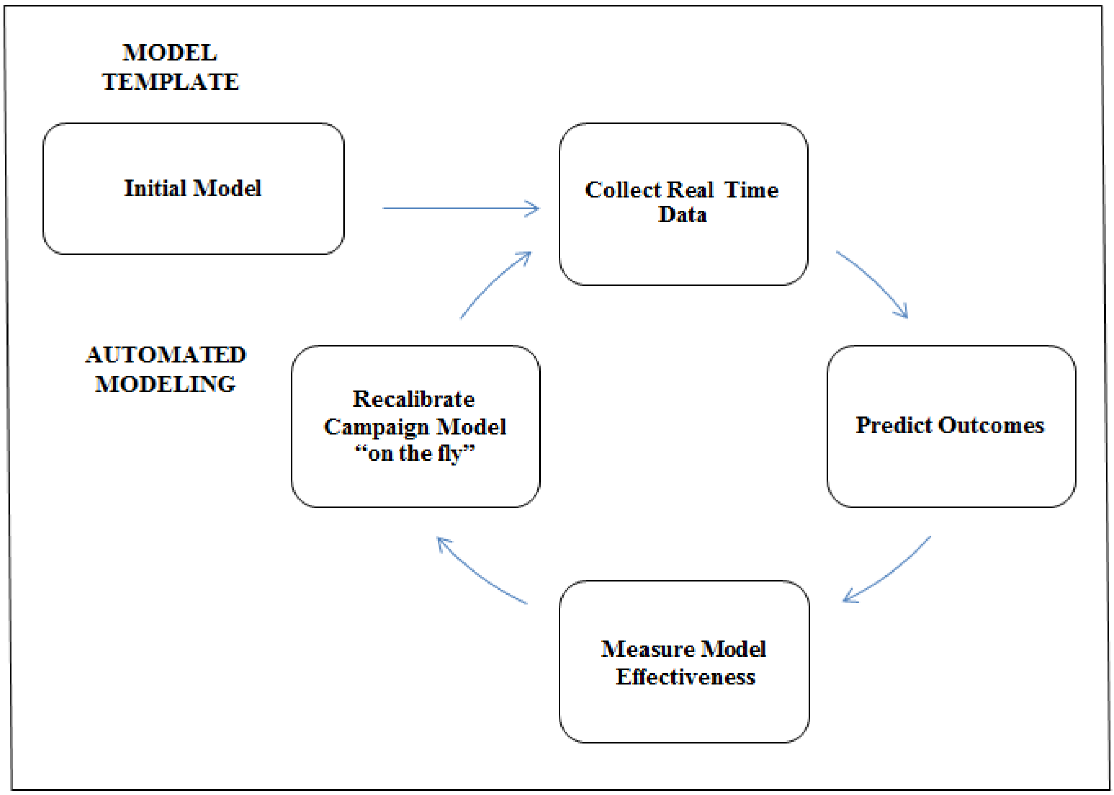

Figure 5. Real Time Analytics Model Feedback Cycle using Automated Modeling

Another dimension of NWDSS decision analytics that does rely upon more traditional model management is the need for computational modeling, especially agent-based modeling and simulation (ABMS), which is an important macro decision technology for generating hypotheses about NWDSS which can be subsequently tested in field experiments and exercises. This capability is also mentioned as a desideratum by the organization science research community for advancing the theory and practice of Transaction Memory Systems (see Table 3).

5. Summary and Conclusions

The NWDSS conceptual model reflects the significantly more dynamic, feedback-driven, and complex nature of decision-making that results from distributed, mobile, collaboration-centric environments. The network nature of contemporary information systems at both social and physical levels has outstripped the ability of traditional DSS precepts and technologies to keep pace. We believe the NWDSS addresses this gap and provides a meaningful foundation for motivating research in this arena. In the above sections we have discussed some of the interesting research issues emerging from NWDSS. Tables 7a-c provide a more detailed, although nowhere near exhaustive, summary of the many issues and challenges in each of the three domains of our NWDSS conceptual model.

Network Infrastructure: Physical Network	
Research Topic	**Research Questions and Related Work**
Adaptive network management	• Theoretical model for network adaptation • Embedding decision support in the physical network (self-aware hyper-nodes for managing physical network [7]); • Disruption tolerant networks: how to maintain continuity in network operations when nodes enter and leave in unanticipated fashion;
Control vs. flexibility in ad hoc networks	Do centralized nodes emerge and if so, how and under what conditions? What tradeoffs exist between control and flexibility and how are these managed? [56]
Tracking and searching in mobility environments	Ontology for searching and tracking • Semantic model for Track [16]; • What are the semantics of missing nodes? What does it mean when a node leaves /enters the network?
Incorporating advanced sensor technology	Unmanned Aerial Vehicles (UAV), Unmanned Surface Vehicles (USV), Unmanned Ground Vehicles (UGV); Sensor-sensor interaction; Scalable sensor data fusion from nano-sensors to satellites
Cyber distortion management	• Error propagation in networks; Interplay between physical and social network error propagation • How to accommodate "lens blur" effects in DSS [21]
Cloud computing	Cloud computing taxonomy and services [5]

Table 7a. NWDSS-Related Research Topics and References for Network Infrastructure

NWDSS Knowledge Mesh: Transactive Memory Systems	
TMS theory and extensions	• Development of organizational-level TMS in the context of social media [31]; • Coordination of emergent knowledge processes, e.g. coordinating situational awareness with individual context awareness; • Alternative organizational knowledge models, e.g. Can organizational memory be modeled as a neural network?
Group / team performance	Identification of valid and consistent TMS measures for NWDSS environments; • Measuring TMS effectiveness, e.g., group performance and decision quality [34]; • Measuring collaborative capacity of virtual teams [19]; • Dynamic identification of weak/ strong links in the social network;
Collaborative engineering	The design of collaborative software and systems • Methodologies for rapid integration of existing systems (system of systems) in near real time [11]; • TMS environments [39; 24] • Open source models for designing and implementing collaborative systems
Expert reach back functionality	• How to identify expertise within a network? [40] • Networks and brain metaphor: Managing sensor-expert interactions via cortex services [8];

Table 7b. NWDSS-Related Research Topics and References for Knowledge Mesh and Transactive Memory Systems

Emergent Decision-Making: Agile Collaboration	
Collaborative DMing models	• Decision-making in adaptive networks Agile collaboration model and requirements [1; 38]; • Explicit-Implicit Interaction theory [17]; • Open source decision-making (e.g., via crowdsourcing) [18; 41]; • Interaction between roles, tasks, and decision processes in NWDSS; Workflow models for decision processes • Function-structure-process model of adaptive decision-making [29]
Decision analytics	Design and implementation of analytical and computational models to work in near real time situations: • Burst analysis [45]; • Word clouds [33]; • Social network analytics, e.g. link analysis [55; 58]; • Agent-based simulation and iterative participatory modeling [4]; • Operations research (e.g., swarms [12]; multi-criteria models [53]); • Visual analytics • Big data [37] ○ Predictive analytics as a service [27]; ○ Real-time predictive analytics and adaptive modeling [28]

Integrating physical and social networks	Comparison of NWDSS frameworks • Multidimensional networks [10; 31] • Digital infrastructure "in the large" as described in [14; 52; 56] • DoD requirements for operational decision making in networked systems of humans and machines [6; 38]
VIRT (Valued Information at the Right Time)	Information filtering for decision-makers in high intensity collaborative environments [15]
Psychology of decision-making	• Incorporating human biases into NWDSS; "slow" vs. "fast" thinking in decision processes [22]; How do individual biases affect collaborative decision-making? What relationships exist between individual and group biases? • Neuroscience foundations of decision-making [23]; • Nano-sensors for monitoring real-time individual decision-making at the biological level
Decision superiority	What are appropriate performance metrics for decision support effectiveness and decision superiority in collaborative environments? How can networks be designed and/or adapted to better achieve those metrics?

Table 7c. NWDSS-Related Research Topics and References for Emergent Decision-Making and Agile Collaboration

We have discussed the convergence of mobile networks, smart handheld devices, social media communications, and semi-autonomous sensors in the context of ad hoc, self-organizing systems and indicated how this substantively changes the nature of decision-making and decision support, resulting in a new kind of information system we call NWDSS. NWDSS is considerably more complex than traditional DSS in the following dimensions: interplay between social and physical networks, human-sensor coordination, feedback-driven emergent processes, dynamic collaboration including expert reach back, and real-time data-driven decision analytics. We've proposed a 3-pillared conceptual model consisting of digital infrastructure, transactive memory systems, and emergent decision-making to encapsulate the critical dimensions of NWDSS environments. Envisioning DSS as networks allows us to extend our thinking about decision support in much broader terms. Conceiving NWDSS as neural networks or cortex memory, for example, opens avenues of exploration into the connections between psychology, neuroscience, brain science and decision-making with the potential to dramatically extend the frontiers of current DSS research.

The MIO exercises we have discussed give rise to palpable examples of decision problems where NWDSS makes a significant difference by providing support that is not possible with standard decision technologies. Adaptive network management, social media, human-autonomous sensor interaction, transactive memory, and emergent collaborative decision-making to name a few are beyond the purview of current traditional DSS capabilities. Our ongoing MIO experiments provide a solid research design foundation for continued generation and testing of NWDSS requirements and functionality. Other representative NWDSS testing scenarios we are considering include casualty and logistics support in disaster relief operations, decision support for manned-unmanned teaming in countering IED threats, and monitoring flash crowds in urban environments.

Although the development of NWDSS concept has been driven by our primary application focus on tactical military operations, we believe that it is sufficiently generalizable to numerous volatile, near-extreme decision-making environments including emergency response, disaster relief, border patrol, search and rescue, widespread social media phenomena, and "on the move" information systems. As such, we argue that NWDSS is a viable launching pad for shaping the next stage in DSS evolution, in which self-organizing mesh network driven decision support and information exchange through highly discrete moments of time and space could take the central stage.

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